



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Experiment 05 of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. This task focuses on Supervised Learning, specifically looking at Regression and Classification tasks. M-Tech Production & Marketing, based in Haripur, KPK, provides this hands-on lab training to teach us how to build predictive models, split datasets into training/testing zones, and evaluate their success using professional metrics.

2. Objectives

- Understand Supervised Learning and its two major categories: Regression and Classification.
- Learn how to split data into training sets and testing sets using `train_test_split`.
- Explore core algorithms like Linear Regression, Logistic Regression, Decision Trees, and KNN.
- Learn how to read model metrics such as R^2 score, RMSE, and Accuracy.

3. Task Description

The task requires implementing and testing various supervised machine learning algorithms using Scikit-Learn. We need to train a regression model to check numerical errors and set up multiple classification models (like KNN, Logistic Regression, and Decision Trees) on the Iris flower dataset. The performance of these models needs to be calculated, compared, and reported clearly.

4. Work Done

I successfully configured and executed all five lab tasks inside a Jupyter Notebook. I generated synthetic numbers to test standard errors, loaded real classification datasets, and compared how changing parameters (like the value of K in KNN) affects prediction results. All results, tables, and accuracy scores were verified through code outputs.

5. Challenges Faced

- Regression vs. Logistic Regression: Understanding that *Linear Regression* is for predicting continuous numbers while *Logistic Regression* is actually used for predicting categories (Classification) was a bit confusing at first.

- **Metric Interpretation:** Learning how to read R^2 (where closer to 1 is better) versus RMSE (where lower is better) takes some practice to understand quickly.

6. Key Learnings

- **Regression vs. Classification:** Regression predicts continuous numbers (like tomorrow's temperature or house prices). Classification predicts classes or labels (like pass/fail or cat/dog).
- **KNN Mechanic:** K-Nearest Neighbors makes predictions by taking a vote from the closest data points. Changing the K value changes how the model votes.
- **Decision Trees:** Decision Trees act like a flowchart asking yes/no questions, making it very easy to see which features are most important for making a choice.

7. Conclusion

This lab provided a clear, practical understanding of supervised machine learning workflows. By training and comparing multiple models, I learned that no single algorithm is perfect for every problem. Evaluating models on unseen test data using clear metrics like accuracy and RMSE is highly important to ensure the AI makes correct decisions in real-world scenarios.

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